

SOC ANALYST LAB REFERENCE

Wazuh Home Lab

Scenario Step Lists

Brief, repeatable workflows organized by attack type

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PURPOSE A concise operational record of each completed and planned SOC scenario. Commands are limited to the isolated, authorized SOC-Lab environment.

Scenario	Status	Primary telemetry
SSH password guessing	Completed	Linux sshd / authentication
SQL injection	Completed	Apache access log
Directory traversal	Completed	Apache access log
Blocked network connection	Completed	UFW / kernel firewall
Failed privilege escalation	Completed	sudo / authentication

Environment: Wazuh 10.10.10.3 • Ubuntu 10.10.10.4 • SOC-Lab 10.10.10.0/24 • Dashboard
<https://127.0.0.1:8443>

1. SSH Password-Guessing Detection

STATUS COMPLETED — Wazuh rule 5710, level 5; Password Guessing and SSH context.

1. Start and verify the lab

Start both VMs and confirm Ubuntu can reach the Wazuh manager.

```
ping -c 4 10.10.10.3
```

2. Configure Wazuh syslog reception

Allow TCP syslog from only the Ubuntu endpoint in /var/ossec/etc/ossec.conf.

```
<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>tcp</protocol>
  <allowed-ips>10.10.10.4</allowed-ips>
  <local_ip>10.10.10.3</local_ip>
</remote>
```

3. Validate and restart Wazuh

Check XML syntax, restart the manager, and verify TCP 514 is listening.

```
sudo systemctl restart wazuh-manager
sudo ss -lntp | grep :514
```

4. Configure Ubuntu forwarding

Forward informational-and-higher syslog messages to Wazuh using TCP.

```
*.info@@10.10.10.3:514
```

5. Generate the controlled event

Use logger to create one uniquely identifiable failed-SSH message.

```
logger -p authpriv.notice -t sshd "Failed password for invalid user soc-lab-test from 10.10.10.99 port 4444 ssh2"
```

6. Investigate in Wazuh

Search Discover for soc-lab-test and record the rule, severity, source, MITRE fields, and full log.

7. Disposition

Classify as a true positive detection with expected benign lab activity; confirm no successful authentication occurred.

2. SQL-Injection Detection

STATUS COMPLETED — Wazuh rule 31103, level 7; MITRE ATT&CK T1190.

1. Install and verify Apache

Install Apache, confirm the service is active, and verify the local default page.

```
sudo apt-get install -y apache2
sudo systemctl is-active apache2
curl -I http://127.0.0.1
```

2. Collect the Apache access log

Configure rsyslog imfile to monitor access.log with an identifiable tag.

```
module(load="imfile")
input(type="imfile" File="/var/log/apache2/access.log" Tag="apache-access:" Severity="info" Facility="local6")
```

3. Validate log forwarding

Check rsyslog syntax, restart it, and confirm it remains active.

```
sudo rsyslogd -N1
sudo systemctl restart rsyslog
sudo systemctl is-active rsyslog
```

4. Create a benign baseline

Request a nonexistent page with a unique user-agent and confirm the 404 access record.

```
curl -A "SOC-Lab-Baseline" http://127.0.0.1/soc-baseline-test
```

5. Generate the controlled SQLi pattern

Send one URL-encoded union/select/from request to a nonexistent local application.

```
curl -A "SOC-Lab-SQLi-Test" "http://127.0.0.1/index.php?id=1%20union%20select%20username,password%20from%20users--"
```

6. Verify the access record

Confirm the marker, URL, and HTTP 404 response in Apache's log.

```
sudo tail -n 3 /var/log/apache2/access.log
```

7. Investigate in Wazuh

Search full_log for SOC-Lab-SQLi-Test and record rule 31103, level 7, decoded HTTP fields, and T1190.

8. Disposition

Document a true positive detection but unsuccessful exploit: the application did not exist and returned 404.

3. Directory-Traversal Detection

STATUS COMPLETED — Wazuh rule 31104, level 6; relevant T1083 and T1190 context.

1. Verify connectivity and services

Confirm Wazuh connectivity, Apache operation, and rsyslog status.

```
ping -c 4 10.10.10.3
sudo systemctl is-active apache2
sudo systemctl is-active rsyslog
curl -I http://127.0.0.1
```

2. Generate one traversal request

Preserve the raw ../ segments and send the request only to localhost.

```
curl --path-as-is -A "SOC-Lab-Path-Traversal" "http://127.0.0.1/../../etc/passwd"
```

3. Apply the safety check

Require a 400, 403, or 404 response. Stop immediately if protected file contents appear.

4. Verify Apache evidence

Confirm the path, HTTP result, and unique marker in access.log.

```
sudo tail -n 3 /var/log/apache2/access.log
```

5. Investigate in Wazuh

Search full_log for SOC-Lab-Path-Traversal and expand the matching event.

6. Record the detection

Capture rule 31104, level 6, Common web attack, source 127.0.0.1, URL, response 400, and MITRE metadata.

7. Interpret mappings carefully

Treat T1083 and T1190 as relevant context; do not claim T1055 process injection without supporting evidence.

8. Disposition

Document a true positive detection with no successful file access or data exposure.

4. Blocked Network Connection

STATUS COMPLETED — Custom Wazuh rule 100100, level 7; ten UFW blocked-connection alerts for TCP/8080.

1. Inspect the current firewall

Confirm UFW is installed and determine whether it is active before changing policy.

```
sudo ufw status verbose
```

2. Preserve approved services

Allow SSH and Apache before enabling the default incoming-deny policy.

```
sudo ufw allow 22/tcp comment 'SOC Lab SSH'
sudo ufw allow 80/tcp comment 'SOC Lab Apache'
```

3. Set policy and logging

Deny unsolicited inbound traffic, allow outbound traffic, enable medium logging, and activate UFW.

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw logging medium
sudo ufw enable
```

4. Create the controlled block

Add an explicit deny rule for the unused test port and verify the numbered rules.

```
sudo ufw deny 8080/tcp comment 'SOC Lab Block Test'
sudo ufw status numbered
```

5. Generate the blocked connection

From Wazuh 10.10.10.3, request TCP/8080 on Ubuntu 10.10.10.4 and confirm a timeout.

```
curl -v --connect-timeout 5 http://10.10.10.4:8080/SOC-Lab-Blocked-Connection
```

6. Verify local firewall evidence

Confirm UFW recorded the source, destination, TCP protocol, SYN flag, and destination port.

```
sudo grep 'DPT=8080' /var/log/ufw.log | tail -n 5
```

7. Confirm syslog transport

Verify rsyslog has an established TCP session to the Wazuh listener.

```
sudo ss -tnp | grep '10.10.10.3:514'
```

8. Create and validate the detection

Back up local_rules.xml, add custom rule 100100 matching UFW BLOCK, validate XML, and test with wazuh-logtest before restarting the manager.

9. Activate and investigate

Restart wazuh-manager, repeat the blocked request, then search Discover for rule.id:100100 and DPT=8080.

10. Verify approved service availability

Confirm Apache still returns HTTP 200 through allowed TCP/80.

```
curl -sS -o /dev/null -w 'Approved Apache service returned HTTP %{http_code}\n' http://10.10.10.4
```

11. Disposition

Document a true positive expected lab event: ten packets blocked, no TCP/8080 session established, and approved Apache access remained available.

5. Failed Privilege-Escalation Attempt

STATUS COMPLETED — Wazuh rule 5404, level 10; MITRE ATT&CK T1548.003 Sudo and Sudo Caching.

1. Record the identity baseline

Confirm the current account, group membership, rsyslog status, and TCP/514 connection.

```
whoami
id
systemctl is-active rsyslog
ss -tn | grep '10.10.10.3:514'
```

2. Clear cached sudo authentication

Invalidate any cached sudo credential so the next request requires authentication.

```
sudo -k
```

3. Generate the controlled failure

Request the harmless `/usr/bin/true` command through sudo and intentionally enter an incorrect password three times.

```
sudo -p 'SOC-Lab-Sudo-Failure: ' /usr/bin/true
```

4. Confirm the terminal result

Verify sudo reports three incorrect password attempts and returns to the unprivileged shell.

5. Verify the local authentication record

Use the correct password for this read-only check and confirm `USER=root`, `COMMAND=/usr/bin/true`, and three failures.

```
sudo grep -E 'authentication failure|incorrect password attempts' /var/log/auth.log | tail -n 10
```

6. Confirm telemetry continuity

Verify rsyslog still has an established connection to Wazuh after the event.

```
ss -tn | grep '10.10.10.3:514'
```

7. Investigate in Wazuh

Search `full_log` for incorrect password attempts, distinguish the failed event from the later `grep` command, and expand the `/usr/bin/true` record.

8. Record the detection

Capture rule 5404, level 10, Three failed attempts to run sudo, source user socstudent, destination user root, and T1548.003.

9. Assess impact and disposition

Confirm `/usr/bin/true` did not execute, no root session or system change occurred, and classify the alert as expected authorized lab activity.

Usage Notes

- Run these steps only in the isolated SOC-Lab environment.
- Take screenshots of the command, local evidence, Wazuh result, and MITRE fields.
- Use a unique marker for every scenario so events are easy to locate.
- After each scenario, record the exact observed rule rather than relying on the expected rule.
- Update planned sections only after the commands and results have been verified.